

OphEdit: Training-Free Text-Guided Editing of Ophthalmic Surgical Video

Ritul Jangir^{*1}, Arkya Jyoti Bagchi^{*1}, Mangalton Okram¹, Saurabh Seetaram Korgaonkar², Aiman Farooq¹, and Deepak Mishra¹

¹ Indian Institute of Technology Jodhpur, 342030 India
`farooq1.0iitj.ac.in, dmishra@iitj.ac.in`

² All India Institute of Medical Sciences Delhi, 110029

Abstract. High-fidelity surgical video generation holds great promise for medical training and AI development, yet precise, controllable editing of such videos remains a formidable challenge. Modifying surgical attributes such as instrument, tissue interactions or procedural phases is particularly difficult due to the strict anatomical and temporal constraints of ophthalmic procedures. We introduce OphEdit, the first training-free framework for text-guided editing of ophthalmic surgical videos. Our approach uses a deterministic second-order ODE inversion pipeline to capture attention Value tensors from the source video and selectively injects them into the conditional branch of classifier-free guidance during denoising, preserving the intricate anatomical geometry of the eye while mapping text-driven semantic modifications onto the video. Clinical evaluation by an ophthalmologist shows that OphEdit handles complex transformations, such as instrument swaps and procedural variations, with superior structural fidelity and temporal consistency compared to adapted general-domain video editors, offering a scalable route to diverse surgical training data. The code and prompts can be accessed at <https://github.com/ophedit/OphEdit>

Keywords: Surgical video editing · Training-free · Diffusion models · Ophthalmic surgery

1 Introduction

Ophthalmic surgical videos are an indispensable resource for medical education, surgical training, and the development of AI-driven assistance systems. Recordings of procedures such as cataract surgery capture fine-grained instrument manoeuvres, tissue responses, and the sequential flow of surgical phases, and recent AI models can already parse them to understand workflow, assess surgeon performance, and guide robotic assistants [10]. Training and validating such systems, however, demands large quantities of annotated video covering diverse surgical variations, rare complications, and subtle deviations in technique.

^{*} Contributed equally

Acquiring surgical video data at this scale remains difficult. Strict patient privacy regulations, ethical constraints, and the high cost of expert annotation limit both the volume and diversity of available data, while clinically critical but infrequent events, such as posterior capsule rupture, are chronically underrepresented. To mitigate this bottleneck, text-guided video generation has emerged as a promising strategy for synthesising surgical content from natural language descriptions [8].

Recent text-to-video (T2V) diffusion models, including CogVideoX [18], HunyuanVideo [6], and Open-Sora [21], achieve impressive realism in natural domains. In the medical domain, Endora [7] and Bora [15] explore generative models for endoscopy and broader biomedical video, while Ophora [8] adapts CogVideoX to ophthalmic surgery through large-scale transfer learning, producing realistic cataract surgery clips from text prompts. These approaches, however, operate by regenerating entire sequences from scratch. This regeneration-based paradigm is computationally expensive and offers little fine-grained control over specific procedural attributes. What clinically constrained settings need instead is the ability to *edit* an existing surgical video, *e.g.*, to swap one instrument for another or alter a single procedural step, while rigorously preserving the delicate anatomical geometry of the eye. Controllable, text-guided editing of ophthalmic surgical videos remains largely unexplored.

Training-based editing frameworks typically require paired data or task-specific fine-tuning, which is impractical in the surgical domain. Training-free methods such as FateZero [11], UniEdit [1], and FLATTEN [4] instead leverage pre-trained diffusion models, but applying them directly to surgical video distorts delicate anatomical structures and violates the strict temporal continuity of instrument motion. To bridge this gap, we propose OphEdit, whose contributions are threefold:

- We present the first training-free video editing framework designed for ophthalmic surgery, enabling precise, language-driven manipulation of procedural elements such as instrument type, tissue appearance, and surgical phase.
- We develop a deterministic second-order ODE inversion scheme that extracts attention Value (V) tensors from a source video and selectively injects them into the conditional branch of classifier-free guidance during denoising, preserving anatomical fidelity while executing text-prompted edits.
- We introduce a second-order midpoint integration strategy for both inversion and generation. By computing dual-pathway latent updates at the log-SNR midpoint, OphEdit eliminates the jittering artifacts of first-order methods, ensuring the smooth procedural dynamics essential for clinical realism.

2 Related Works

2.1 Natural Video Editing

Video editing through textual data has been progressing in different directions, from zero-shot approaches to tuning a ViT using 2D images. Zero-shot approaches using pretrained diffusion models for editing without retraining or

per-video optimization have been achieved by FateZero [11], UniEdit [1], and FLATTEN [4].

FateZero [11] achieves this by storing self- and cross-attention maps during DDIM inversion, retaining structural and motion information, and reinjecting them during denoising while minimizing semantic leakage. UniEdit [1] extends the tuning-free approach beyond appearance to motion editing, introducing separate reconstruction and motion-reference branches that inject spatial and temporal self-attention into the main editing path, so appearance and motion can be controlled largely independently. FLATTEN [4] improves frame consistency by incorporating optical flow directly into the attention module, constraining patches along the same flow trajectory to attend to each other so the same physical point stays consistent across frames.

SAMA [20] factorizes instruction-guided video editing into Semantic Anchoring – jointly predicting semantic tokens and video latents at sparse anchor frames for instruction-aware structural planning – and Motion Alignment, which pre-trains the same backbone on motion-centric restoration tasks (cube inpainting, speed perturbation, tube shuffle) so temporal dynamics are learned directly from raw video rather than injected externally. EditVerse [5] unifies image/video generation and editing as a single token-sequence model, using full self-attention for in-context learning across tasks and modalities, which lets it transfer editing knowledge from abundant image data into video. ViFeEdit [19] targets the video-supervision bottleneck by decoupling spatial from temporal token modeling in a DiT, so the spatial-editing parameters can be tuned purely on 2D image pairs while the base model’s pretrained temporal attention stays untouched, preserving motion coherence without any video training data.

2.2 Surgical Video Generation

Text-guided end to end synthesis methods like Ophora [8] and SurGen [3] are built on top of already existing T2V models rather than building surgery-specific architecture. Ophora, built on Ophora160K dataset of video-instruction pairs, specifically is designed for ophthalmic surgical video generation, using progressive video-instruction tuning to transfer spatiotemporal knowledge from natural domain T2V model to ophthalmic surgical domain. SurGen [3] adapts CogVideoX [18] architecture to laparoscopic cholecystectomy, generating surgical videos with higher resolution and longer duration than prior works which were conditioned only on coarse phase-level text without structural intermediates.

Beyond text conditioning, a second line of work injects structured intermediates — segmentation maps, scene graphs, or symbolic rules — to gain finer control over anatomy and instrument placement than free-form prompts allow. HieraSurg [2] decomposes generation hierarchically: a first diffusion stage predicts panoptic segmentation maps from surgical phase and action-triplet labels, and a second stage renders video conditioned on these maps, giving explicit control over anatomy and instrument layout that pure text conditioning lacks.

SG2VID [12] pushes controllability further by conditioning on scene graphs instead of segmentation maps, letting a user directly edit entity size, motion, or the entrance of new tools — control that HieraSurg [2]’s dense maps do not expose as easily. The most recent of these structured-conditioning approaches, SWoMo [13], pushes control a step further by separating what happens from what it looks like: a symbolic component — a rule-based simulator paired with scene graphs — governs tool-tissue motion, while a separate diffusion component is responsible only for rendering visual realism. This split keeps motion physically grounded even for tool trajectories that fall outside the training distribution, something purely learned motion models struggle with, marking a shift toward decoupling physical plausibility from visual synthesis in surgical simulation.

Data scarcity itself has also been treated as a first-class problem rather than a side effect. SurV-Gen [17] addresses this directly with a lightweight two-stage diffusion design that decouples spatial from temporal modeling, allowing it to reuse pretrained image-diffusion weights and generate convincing clips of underrepresented surgical actions or instruments from only a handful of examples — footage that then improves downstream recognition models trained on the augmented data. Whether conditioned on text, structured priors, or optimized for data efficiency, these methods share a common goal: producing new surgical footage from scratch. None of them address editing an existing surgical video while keeping its underlying anatomy and instrument identity intact — the gap OphEdit fills.

3 Methodology

OphEdit is a training-free framework for text-guided editing of ophthalmic surgical videos. Figure 1 shows the overall pipeline: (1) deterministic inversion of a source video to extract and store structural attention tensors, and (2) text-guided denoising from the inverted noise, where the stored tensors are selectively injected to preserve anatomy while applying the semantic edit.

3.1 Backbone: CogVideoX and Ophora

OphEdit builds on CogVideoX-2B [18], a diffusion transformer for text-to-video generation, and its ophthalmic-domain variant Ophora [8]. Video frames are compressed by a 3D causal VAE ($8\times$ spatial, $4\times$ temporal) into a latent $z \in \mathbb{R}^{T \times H \times W \times C}$; text prompts are encoded by a frozen T5-XXL encoder and concatenated with video tokens along the sequence dimension. CogVideoX employs joint 3D full attention, so every token attends to every other token across all frames and patches, rather than factorising attention into separate spatial and temporal blocks; this is what lets our editing intervention target the attention Value (V) tensors directly, since every frame’s content passes through the same joint attention operation. Ophora adapts this backbone to ophthalmic surgery via a two-stage fine-tuning process on a large corpus of narrated cataract-surgery

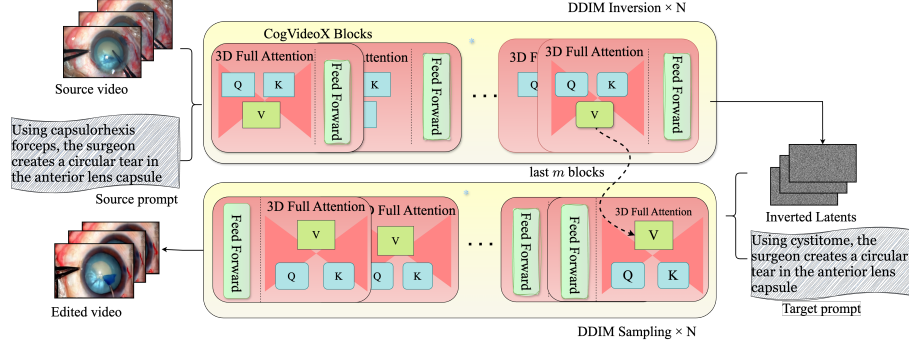


Fig. 1: Overview of our proposed training-free video editing framework. In the inversion stage, second-order DDIM inversion captures the attention Value tensor at both the full-step and midpoint evaluations. In the denoising stage, the reverse process is guided by the target prompt with timestep-aligned injection of the stored values.

clips: transfer pre-training instils general ophthalmic domain knowledge (instrument appearances, anatomical structures, phase-specific dynamics), followed by fine-tuning on a privacy-filtered subset to align the output distribution with clinical privacy standards. The resulting model generates high-fidelity, anatomically plausible surgical videos conditioned on detailed text prompts. OphEdit uses this frozen Ophora model both to generate source videos and as the backbone for editing; no parameters are updated at any stage of our pipeline.

3.2 Deterministic Inversion with Second-Order Integration

Given a source video V generated by Ophora from prompt p_s , we encode it into latent space via the VAE to obtain z_0 . The goal of inversion is to recover a noise latent z_T such that denoising from z_T with p_s reconstructs z_0 . Standard DDIM inversion [14] provides a deterministic forward process, advancing the latent at each step as

$$z_{t_{i+1}} = \sqrt{\bar{\alpha}_{t_{i+1}}} \hat{z}_0 + \sqrt{1 - \bar{\alpha}_{t_{i+1}}} \hat{\epsilon}_\theta(z_{t_i}, t_i, p_s), \quad (1)$$

where $\hat{\epsilon}_\theta$ is the model’s noise prediction, $\bar{\alpha}_t$ is the cumulative noise schedule, and $\hat{z}_0$ is the clean latent estimate derived from z_{t_i} and $\hat{\epsilon}_\theta$. This is an Euler discretization of the underlying probability flow ODE, with local truncation error $O(\Delta t^2)$ that accumulates across the trajectory; in ophthalmic surgical video this manifests as high-frequency jitter that violates the continuous physics of instrument motion.

Why second-order integration. Simply increasing the number of first-order steps only reduces this error linearly, and matching a second-order solver’s $O(\Delta t^3)$ accuracy this way costs roughly twice the neural function evaluations (NFEs) of running a genuine predictor-corrector step at the original step count. Adaptive solvers such as DPM-Solver++ [9] reach second-order accuracy efficiently for sampling, but rely on non-fixed, adaptively chosen intermediate

timesteps; since our editing stage keys stored attention tensors to specific timesteps and must inject them at those same steps during denoising, this adaptive scheduling is incompatible with our requirements. We therefore adopt a fixed-schedule second-order midpoint scheme (Heun’s method), which reaches the same asymptotic accuracy at equal compute while preserving exact timestep alignment between inversion and editing.

For each macro-step $t_i \rightarrow t_{i+1}$, we perform: (1) a *predictor* pass that computes the full-step noise prediction $\hat{\epsilon}_{\text{full}} = \hat{\epsilon}_\theta(z_{t_i}, t_i, p_s)$ and its clean latent estimate $\hat{z}_0^{\text{full}}$, then takes a temporary Euler step to the log-SNR midpoint t_{mid} to obtain an intermediate state z_{mid} ; (2) a *corrector* pass that evaluates the slope at this intermediate state, $\hat{\epsilon}_{\text{mid}} = \hat{\epsilon}_\theta(z_{\text{mid}}, t_{\text{mid}}, p_s)$ and $\hat{z}_0^{\text{mid}}$; and (3) an *averaging* step, where $\hat{z}_0^{\text{avg}} = (\hat{z}_0^{\text{full}} + \hat{z}_0^{\text{mid}})/2$ and $\hat{\epsilon}^{\text{avg}} = (\hat{\epsilon}_{\text{full}} + \hat{\epsilon}_{\text{mid}})/2$ are substituted into the DDIM update in Eq. (1) to reach $z_{t_{i+1}}$.

The midpoint timestep is chosen in the linearised log-SNR space rather than by simple index averaging, since the DDPM schedule is non-linear and $(t_i + t_{i+1})/2$ would not correspond to the physical midpoint of the noise transition:

$$\lambda(t) = \frac{1}{2} \log \left(\frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t} \right), \quad t_{\text{mid}} = \arg \min_k \left| \lambda(k) - \frac{1}{2}(\lambda(t_i) + \lambda(t_{i+1})) \right|, \quad (2)$$

snapped to the nearest available integer timestep, since the transformer cannot process fractional timesteps without falling into out-of-distribution states.

Attention storage. During both the full-step and midpoint evaluations, we extract the attention Value tensors from the final m transformer blocks, the blocks operating at the highest spatial resolution, where the model captures fine-grained surgical detail such as instrument boundaries and delicate tissue textures, and store them in an attention dictionary keyed jointly by timestep and sub-pass: $S[t_{i+1}]_{\text{full}}$ for the predictor pass and $S[t_{i+1}]_{\text{mid}}$ for the corrector pass. Storage is restricted to the final n inversion steps, the high-noise region in which the global structural layout of the surgical scene is established. We store only the Value tensors, not the Query or Key projections: Q and K primarily govern attention routing, whereas V carries the content, the anatomical structures and textures of the eye, that we want to preserve.

3.3 Timestep-Aligned Value Injection for Editing

To generate the edited video V' , we initialise the reverse diffusion trajectory from the inverted noise latent z_T , conditioned on the target prompt p_t , and run the identical second-order midpoint scheme so that the inversion and editing trajectories remain symmetric and the stored keys stay aligned throughout.

CFG branch isolation. Classifier-free guidance (CFG) computes an unconditional and a conditional noise estimate at each step from two separate forward passes. This separation gives us a critical degree of freedom: we can intervene in the conditional branch alone. Let V_{stored} denote the attention Value tensor retrieved from $S[t]$ (the full-step or midpoint key, matching the current sub-pass) at timestep t . We replace the conditional branch’s own V tensor with V_{stored} in

the final m transformer blocks, while its Query and Key projections, still computed from the target prompt p_t , and the entire unconditional branch are left untouched:

$$\hat{\epsilon}_\theta = \hat{\epsilon}_{\text{uncond}} + \omega(\hat{\epsilon}_{\text{cond}}(V_{\text{stored}}) - \hat{\epsilon}_{\text{uncond}}), \quad (3)$$

where ω is the guidance scale. Because Q and K in the conditional branch continue to be computed from p_t , the model still attends according to the target prompt’s semantics; only the content being attended to (V) is anchored to the source anatomy. This has two consequences. First, *anatomical preservation*: the unconditional branch, being unmodified, continues to represent the marginal distribution learned from the source video and anchors the spatial layout and global anatomy of the eye. Second, *semantic editing*: the conditional branch is forced to follow p_t , but its Value tensors remain constrained to the source geometry, so the guidance term $\omega(\hat{\epsilon}_{\text{cond}} - \hat{\epsilon}_{\text{uncond}})$ drives the sample toward the region where both the target prompt and the original structure are simultaneously satisfied.

Injection schedule. We confine injection to a binary schedule over the first n_{inject} denoising steps, the high-noise steps that govern global structure and spatial composition:

$$\mathcal{I}(i) = 1_{\{i \leq n_{\text{inject}}\}} = \begin{cases} 1, & \text{if } i \leq n_{\text{inject}}, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

When $\mathcal{I}(i) = 0$, the model denoises freely under p_t , refining semantic detail and fine texture without further structural constraint. When $\mathcal{I}(i) = 1$, we intervene in the joint attention layers of the final m blocks to enforce structural alignment between the source and target latents.

Second-order denoising with injection. At each timestep t , the injection is executed twice: a full-step update at (z_t, t) using the full-step lookup key $S[t]_{\text{full}}$ anchors the initial trajectory, followed by a midpoint update at the intermediate latent z_{mid} and its log-SNR midpoint t_{mid} , using the midpoint lookup key $S[t]_{\text{mid}}$ to refine the motion estimate. The final latent z_{t-1} is derived by averaging the two sub-step predictions, mirroring the averaging step used during inversion. This symmetrical formulation maintains a consistent ODE trajectory [14] across inversion and editing, eliminating the flickering artifacts of first-order integration and preserving the continuous procedural dynamics essential for clinical realism. The edited latent z'_0 is decoded through the pre-trained spatio-temporal VAE decoder $\mathcal{D}$ (Section 3.1), where $V' = \mathcal{D}(z'_0)$ encapsulates the textually guided edits while retaining the source video’s anatomical and temporal priors.

4 Experiment and Results

4.1 Dataset and Implementation Details

We evaluate OphEdit using the Ophora-28K dataset [8], a privacy-preserving subset of the larger Ophora-160K collection. Ophora provides a small set of 12

canonical prompts, one per procedural phase, spanning Anterior Chamber Flushing, Capsule Polishing, Capsulorhexis, Hydrodissection, Incision, Irrigation-Aspiration, Lens Implantation, Lens Positioning, Phacoemulsification, Tonifying Antibiotics, Viscoelastic Injection, and Viscoelastic Suction, which are used to condition the pre-trained Ophora model for source video generation. We treat each canonical prompt as a source prompt p_s and generate the corresponding source video with the pre-trained Ophora text-to-video model; since OphEdit is fully training-free, none of these prompts or videos are used for fine-tuning.

To construct editing targets, an expert ophthalmologist modified each of the 12 base prompts to introduce a specific, clinically meaningful procedural change while preserving the underlying anatomical context, for example, replacing a phacoemulsification handpiece with an irrigation-aspiration handpiece, or substituting sodium hyaluronate with trypan blue dye during viscoelastic injection.

Generated clips have an average duration of 5.54 seconds, are uniformly sampled to 49 frames at 16 fps, and are resized to 240×352 before processing. Our framework is implemented in PyTorch, built on the CogVideoX and Ophora codebases. We use $N = 150$ inference steps, inject stored attention values during the first n_{inject} denoising steps (tuned per target prompt; $n_{\text{inject}} = 10$ by default across our main evaluation set), restrict injection to the final $m = 8$ transformer blocks, and use a guidance scale of $\omega = 7.5$. All experiments were run on NVIDIA 80GB A100 GPUs.

4.2 Evaluation Protocol

Text-guided editing of ophthalmic surgical videos has no established benchmark. Therefore, we adopt a clinical evaluation protocol conducted by an experienced ophthalmologist who is blinded to the model identity. For each edited video V' , the evaluator watches the clip alongside its target prompt p_t and assesses how well the video depicts the intended surgical step, action, and clinical intent specified in the prompt, rather than merely whether superficial visual details coincide. Each edited video is evaluated according to the following three criteria using a 3-point scale (0 = completely incorrect, 1 = partially correct, 2 = completely correct):

- **Anatomical Consistency:** Assesses whether the delicate ocular structures (e.g., the cornea, iris, and lens capsule) remain anatomically plausible and free from distortions after editing.
- **Correct Operation:** Assesses whether the intended semantic edit (e.g., instrument replacement or dye substitution) is faithfully performed according to the target prompt.
- **Overall Edit:** Provides a holistic assessment of the edited video by jointly considering anatomical consistency and semantic accuracy.

As baselines, we compare against FateZero [11] and UniEdit [1], both state-of-the-art training-free video editing methods originally designed for natural scenes, adapted to the surgical domain using the same Ophora backbone and

Table 1: Comparison of correctness and editing performance reported on a 0–1 scale. Best values in each column are shown in bold.

Model	Anat.	Op.	Edit
FateZero [11]	1.00	0.29	0.63
FateZero-v2	0.95	0.34	0.53
UniEdit [1]	0.68	0.55	0.63
OphEdit (DDIM)	1.00	0.47	0.63
OphEdit (2nd Order)	1.00	0.55	0.66

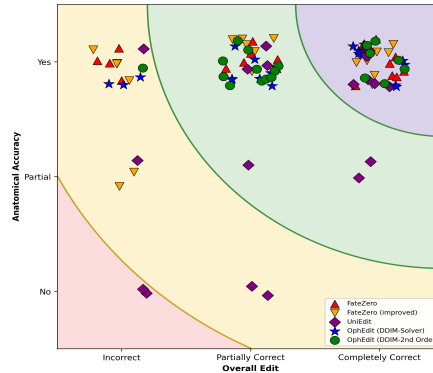


Fig. 2: Distribution of clinical evaluation scores across phase-specific surgical interventions.

the same target prompts as OphEdit. FateZero-v2 restricts attention-map injection to the same fixed final m transformer blocks used by OphEdit, rather than injecting across all blocks as in the original FateZero, isolating the effect of our block-selection strategy when applied to FateZero’s own attention-based editing mechanism. We compare two variants of OphEdit: standard first-order DDIM integration (OphEdit (DDIM)) and our proposed second-order midpoint scheme (OphEdit (2nd Order)).

4.3 Results

Quantitative results are presented in Table 1. OphEdit (DDIM) and OphEdit (2nd Order) both achieve the highest Anatomical Consistency (1.00), matched by FateZero, while UniEdit trails at 0.68 and FateZero-v2 reaches 0.95. On Correct Operation, OphEdit (2nd Order) attains the highest score (0.55), tied with UniEdit, ahead of OphEdit (DDIM) (0.47), FateZero-v2 (0.34), and FateZero (0.29). On Overall Edit, OphEdit (2nd Order) achieves the best score (0.66), narrowly ahead of FateZero, UniEdit, and OphEdit (DDIM), which are tied at 0.63, with FateZero-v2 lowest at 0.53. Figure 2 visualizes the distribution of clinical evaluation scores across all 19 phase-specific interventions. Each point represents a single edited video; the colored regions demarcate performance zones: purple and green indicate successful edits with preserved anatomy, while yellow and red indicate partial or complete failures. OphEdit (2nd Order) has no points falling in the anatomically incorrect row, indicating that it never produces a complete loss of anatomical structure, even on edits that are only partially correct, demonstrating its ability to perform accurate surgical modifications while preserving delicate ocular structures.

To complement the clinical evaluation with an automatic, reproducible metric, we additionally report Fréchet Video Distance (FVD) [16] between each

method’s edited videos and the distribution of real Ophora source videos, shown in Table 2. OphEdit (DDIM) achieves the lowest FVD among all edited-video methods (1388), the closest to the source-video distribution (1192), indicating that its edits remain the most visually and temporally consistent with real ophthalmic surgical footage. FateZero [11] and FateZero-v2 follow (1483 and 1512, respectively), while UniEdit [1] records the highest FVD (1556). Notably, OphEdit (2nd Order) records a higher FVD (1550) than its first-order counterpart, despite achieving the strongest clinical scores in Table 1. We attribute this to the second-order scheme allowing the target prompt to exert a stronger influence on the denoising trajectory, which improves clinically judged edit correctness but introduces frame-level appearance shifts that FVD, which measures distributional similarity to real videos rather than semantic or anatomical correctness, penalises. This divergence between FVD and clinical scoring is itself informative: it suggests FVD alone is an unreliable proxy for editing quality in this domain, and motivates our reliance on clinical evaluation as the primary measure of success, with FVD reported as a complementary, reproducible reference point rather than a substitute.

Table 2: FVD comparison across models. Best values in each column are shown in bold.

Model	FVD
Ophora (source videos)	1192
FateZero [11]	1483
FateZero-v2	1512
UniEdit [1]	1556
OphEdit (DDIM)	1388
OphEdit (2nd Order)	1550

4.4 Qualitative Results

Figure 3 presents a side-by-side comparison of edited video frames produced by OphEdit, FateZero and UniEdit for two representative editing tasks. The top row shows an instrument swap: the source video depicts phacoemulsification with an ultrasonic probe, and the target prompt requests replacement with a coaxial irrigation-aspiration handpiece. OphEdit generates the correct instrument while preserving the surrounding ocular anatomy; FateZero fails to modify the instrument and introduces visual artifacts. The bottom row illustrates a dye substitution: the source video shows injection of clear sodium hyaluronate viscoelastic, and the target prompt requests trypan blue dye. OphEdit renders the blue tint of the dye within the anterior chamber while maintaining crisp anatomical boundaries; FateZero produces inconsistent coloration and blurs the delicate tissue structures.

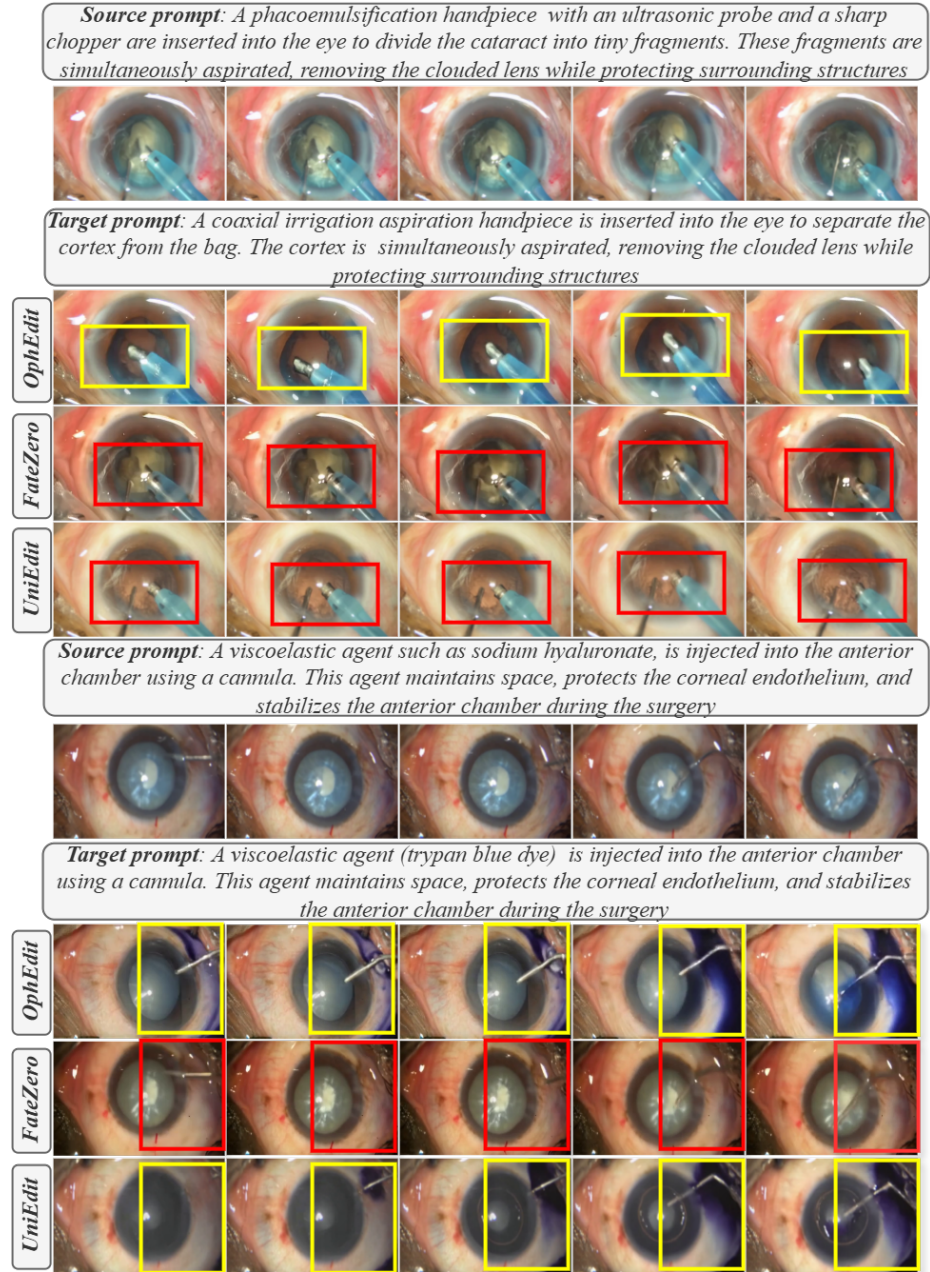


Fig. 3: Qualitative comparison of text-guided surgical video editing between OphEdit, FateZero and UniEdit . Top: editing from Phacoemulsification (source) to Irrigation/Aspiration (target), where the ultrasonic probe is replaced with a coaxial irrigation-aspiration handpiece. Bottom: editing from Viscoelastic Injection (source) to Trypan Blue Dye Injection (target). OphEdit preserves anatomical structures and generates clinically plausible instrument transitions, while FateZero and UniEdit produces visible artifacts and fails to maintain temporal consistency. Yellow boxes indicate correct edits; red indicates incorrect ones.

4.5 Ablation Study

To identify which attention tensor carries the anatomical structure needed for faithful edits, we ablate the choice of tensor(s) injected into the conditional CFG branch during editing (scoped to the last $n=14$ timesteps, last $m=8$ blocks; all else fixed as in Sec. 4.1). Table 3 shows K -only injection barely improves on the uninjected baseline, while V -only injection reaches 1.00 Anatomical Consistency and superior Edit quality (0.63 vs. 0.11). This confirms V encodes the anatomical content that must be preserved, while Q/K mainly govern attention routing — supporting our choice to inject only V .

Table 3: Ablation on which attention tensor(s) are injected into the conditional CFG branch during editing, evaluated on the same 19 clinical edit pairs used in Table 1, reported on a 0–1 scale. Best values in each column are shown in **bold**.

Injected tensor(s)	Anat.	Op.	Edit
None (direct generation)	0.11	0.47	0.11
K only	0.13	0.47	0.11
V only (OphEdit(DDIM))	1.00	0.47	0.63

4.6 Discussion

These qualitative examples, together with the quantitative comparison in Table 1, point to two design choices behind OphEdit’s behaviour. First, selective Value tensor injection, storing and re-injecting only the V tensors rather than full attention maps or the Q, K projections, is intended to balance structural preservation against editing flexibility; we treat this as a design hypothesis motivated by the role of V in encoding content rather than attention routing (Section 3.2), rather than as an independently validated finding. Second, the second-order midpoint integration scheme is designed to remove the high-frequency jitter that first-order solvers introduce, which we expect to be reflected in fewer low-similarity outliers and more temporally coherent instrument motion in the edited videos. However, the clinical evaluation also highlighted the inherent challenges; edits requiring complex fluid-tissue interactions, such as Hydrodissection or deep Phacoemulsification, frequently resulted in lower edit quality scores, as modifying the target prompt occasionally disrupted the temporal logic of the surgical sequence. Despite such cases, our method significantly outperformed the baseline by ensuring that even when an operation failed to execute perfectly, the underlying patient anatomy remained largely intact and free of catastrophic generative artifacts, demonstrating its potential for safe and reliable medical data augmentation.

5 Conclusion

This paper presents OphEdit, the first fully training-free framework for the text-guided editing of ophthalmic surgical videos. To address the strict anatomical and temporal constraints of the medical domain, we propose a novel inversion-injection pipeline built upon the Ophora foundation model. By capturing Attention Value (V) tensors during deterministic DDIM inversion and selectively injecting them into the conditional Classifier-Free Guidance (CFG) branch, our approach rigorously preserves delicate ocular structures while accurately executing text-driven semantic modifications. Comprehensive evaluations by an ophthalmologist demonstrate that OphEdit successfully performs complex surgical transformations, such as instrument substitutions and phase transitions, with superior structural fidelity compared to adapted general-domain baselines.

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